INCH-POUND

MIL-I-27667C(USAF)
16 November 1990
SUPERSEDING
MIL-I-27667B(USAF)
5 March 1969

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

* 1.1 Scope. This specification covers the requirements for a multifunction electrical-resistance temperature indicators.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

EHU-16A/A	White lighted
EHU-29/A	Red lighted

* 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-676	Boxes Set-up

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Resources and Logistics Services Division, SA-ALC/TIRDM, Kelly AFB, TX 78241-5000 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

	MIL-P-116	Preservation, Methods of
*	MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
	MIL-I-7070	Indicator, Temperature, Electrical-Resistance, Hermetically Sealed, General Specification for
	MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
	MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification for
	MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
	MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

FEDERAL

	FED-STD-595	Color

MILITARY

	MIL-STD-129	Marking for Shipment and Storage
*	MIL-STD-130	Identification Marking of U.S. Military Property
	MIL-STD-810	Environmental Test Methods and Engineering Guidlines
	MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
	MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
	MS33678	Connector, Receptable, Electric, Integral Mounting

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto, if applicable.

FEDERAL AVIATION AGENCY

TECHNICAL STANDARD ORDER

	TSO-C43	Temperature Indicators

(Copies of the FAA document may be obtained from the Federal Aviation Agency, Washington, D.C. 20553.)

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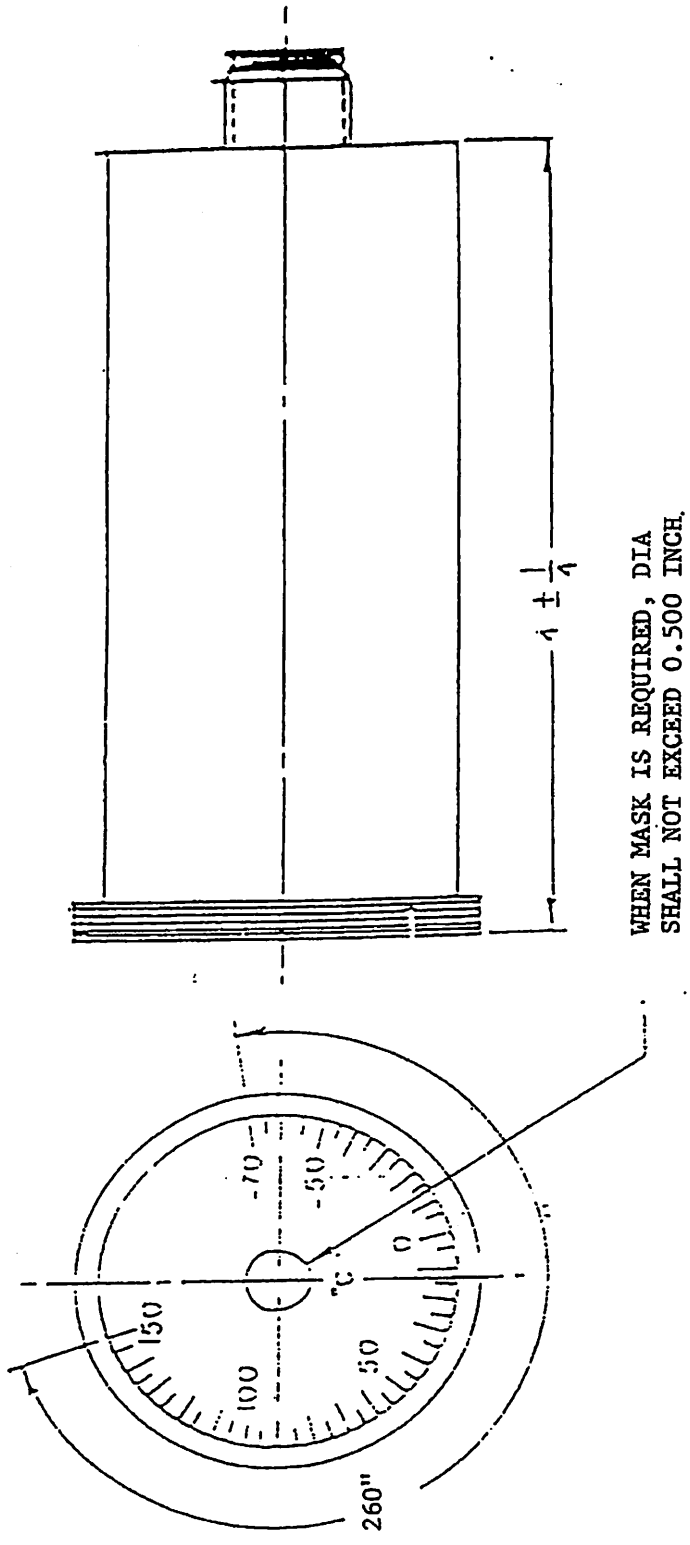


FIGURE 1. INDICATOR DIAL AND CASE

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. Three indicators shall be subjected to all the tests specified in section 4.

3.2 Design and construction.

3.3.1 Case. The length of the case shall not exceed 2-inch diameter size.

* 3.3.1.1 Filling medium. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium, at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The absolute pressure of the filling medium shall be approximately 1 atmoshphere. Leakage shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

3.3.2 Dial. The dial shall conform to figure 1. The dial diameter, as measured across the outer ends of the graduations, shall be not less than 1-3/4 inches. Th 90° indication shall be positioned on the horizontal centerline at the left side of the dial. The dial shall be integrally lighted in accordance with MIL-L-25467 for red lighting and MIL-L-27160 for white lighting, as applicable. The dial blank shall consist of a flat disc of plastic, metal, or other material that will permit compliance with the lighting and other requirements specified herein.

3.3.2.1 Scale arc. The dial shall have a minimum scale arc angle of 260°. The mechanism shall not be geared to obtain the specified scale arc. Adjacent scale graduations shall be spaced within one angular degree of precise linear spacing.

3.3.3 Pointer. The pointer width 0.50 ±0.063 inch from the tip shall be 0.078 ±0.016 inch and shall taper to width of 0.020 ±0.005 inch at the tip. The pointer shall be of such length that the tip will extend 0.019 ±0.016 inch into the shortest graduation. The pointer shall be lighted in accordance with MIL-L-25467 or MIL-L-27160, as applicable.

3.3.3.1 Pointer overswing. The pointer overswing shall be not less than three angular degrees beyond each end of the scale.

* 3.3.4 Cover glass. The cover glass shall be finished with a reflection-reducing coating which shall meet the requirements of MIL-C-14806.

3.3.5 Dial to cover glass distance. The distance between the inner surface of the cover glass and the dial shall not exceed 0.188 inch.

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* 3.3.6 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of 2000 hours demonstrated to a 90 percent confidence level with a discrimination of ratio of 1.5.

3.3.7 TSO conformance. The indicator shall conform to the requirements of Federal Aviation Agency TSO-C43 for type I indicators.

3.4 Power consumption. The indicator power consumption, exclusive of the lighting circuit, shall not exceed 2 watts.

3.5 Lighting. Lighting shall be in accordance with MIL-L-25467 or MIL-L-27160, as applicable.

3.6 Recycled, reclaimed, and virgin materials. Except when intended use of the item will be jeopardized by the use of reclaimed or recycled materials, in preparing new and revising or amending specifications, preparing activities will ensure that:

a. There is no exclusion to the use of recovered materials.

b. There is no requirement that an item be manufactured from virgin materials.

c. Within 1 year from the date of issue by the Environmental Protection Agency of guidelines designating items which are or can be produced with recovered materials, specifications for such products require the use of recovered materials to the maximum extent possible.

3.7 Electrical receptacle. The MS33678-14S-5P hermetically sealed, electrical receptacle shall be located on the vertical and horizontal centerlines and shall be wired as follows. The polarizing pin shall be located at the top center of the receptacle.

PIN	CONNECTION
A	+28 V DC
B	Resistance bulb conecton
C	Ground and remaining bulb connection
D	5V lighting input
E	5V lighting ground

3.8 Electronic components. Electronic components and the application shall conform to MIL-E-5400.

3.9 Jewel bearings. When jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.10 Markings.

3.10.1 Dimensions and colors. The dimensions and colors of the markings shall be as follows. The colors shall be in accordance with FED-STD-595.

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TABLE I

TEST TOLERANCES

Test Point	Test Resistance	Scale Error at Room Temperature	Reduced Voltage	Scale Error at Low Temperature	Scale Error at High Temperature	Scale Error After Vibration
°C	Ohms	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C
-70	60.27	4.4 (5.0)	3.0 (3.0	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
-30	80.56	4.4 (4.4)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
0	90.38	4.4 (4.4)	2.0 (2.0	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
+30	100.91	4.4 (4.4)	2.0 (2.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
80	120.36	4.4 (4.4)	2.0 (2.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
120	137.78	4.4 (4.4)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)
150	151.91	4.4 (5.0)	3.0 (3.0)	6.6 (6.6)	6.6 (6.6)	6.6 (6.6)

Test Point	Test Resistance	Scale Error at Room Temp after Open and Short Circuit	Scale Error with Bulb Lead Tolerance	EXTREME TEMPERATURE EXPOSURE		
				Scale Error at Low Temperature	Scale Error at Room Temp after Low Temperature	Scale Error at Room Temp after High Temperature
°C	Ohms	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C
-70	60.27	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)
-30	80.56	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)
0	90.38	6.6 (6.6)	3.0 (3.0)	7.5 (6.6)	6.6 (4.4)	6.6 (4.4)
+30	100.91	6.6 (6.6)	3.0 (3.0)	7.5 (6.6)	6.6 (4.4)	6.6 (4.4)
80	120.36	6.6 (6.6)	3.0 (3.0)	7.5 (6.6)	6.6 (4.4)	6.6 (4.4)
120	137.78	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)
150	151.91	6.6 (6.6)	3.0 (3.0)	9.5 (6.6)	6.6 (4.4)	6.6 (4.4)

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Marking	Height or Length Inch +0.005	Width of Line or Graduation Inch +0.005	Color No.
Numerals	0.156	0.031	37875
Major graduations	0.156	0.031	37875
Minor graduations	0.094	0.020	37875
Lettering C	0.125	0.020	37875
Background of dial and mask	-----	-----	37038

3.11 Weight. The weight of the indicator shall not exceed 1.0 pound.

3.12 Nameplate. Where applicable, the information on the nameplate shall be in accordance with MIL-STD-130. The nameplate shall be made of metal and shall be permanently attached on the rear of the case.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-7070 with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

* 4.2.1 Preproduction tests. The preproduction test shall consist of all the tests specified herein and in MIL-I-7070.

4.3 Test conditions.

4.3.1 Test voltage. Unless otherwise specified, the test voltage shall be as 28.5 ±0.5V dc and the lighting circuit shall be energized with 4.5 ±0.5V during all tests.

4.3.2 Test tolerances. The tolerances specified in Table I not enclosed in parentheses are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I enclosed in parentheses are applicable only when the indicators are subjected to a specific test independent of sequence.

4.4 Test methods.

4.4.1 Individual tests.

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4.4.1.1 Scale error at room temperature. The test resistance specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in Table I shall not exceed the tolerance specified in Table I.

4.4.1.2 Position error. The position error of the indicator shall not exceed 3°C.

4.4.1.3 Friction error. The friction error of the indicator shall not exceed 2°C.

4.4.1.4 Insulation leakage. Insulation resistance shall be a minimum of 20 megohms.

4.4.1.5 Lighting. The indicator shall be subjected to the individual lighting tests of MIL-L-25467 or MIL-L-27160, as applicable.

4.4.1.6 Sealing. The indicator shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.4.2 Sampling plan A.

4.4.2.1 Reduced voltage operation. The indicator shall be subjected to the scale error at room temperature test within an applied voltage of 22.5V at the test points specified in Table I. The indications at the reduced voltage shall not differ from the indications with 28.5V applied by more than the tolerances specified in Table I.

4.4.2.2 Scale error at low temperature. The scale error shall not exceed the tolerances specified in Table I for low temperature.

4.4.2.3 Scale error at high temperature. The scale error shall not exceed the tolerances specified in Table I for high temperature.

4.4.2.4 Magnetic effect. The magnetic compass shall not deflect more than 5°.

4.4.2.5 Resistance bulb current. A total resistance of 100 ohms shall be applied across the resistance bulb terminals of the indicator, the test voltage applied, and the resistance bulb circuit current measured. The current shall not exceed 22 ma.

4.4.2.6 Open and short circuit. After open and short circuit, the scale errors shall not exceed the tolerances specified in Table I.

4.4.2.7 Vibration error. The indicator shall be subjected to a random vibration test in accordance with MIL-STD-810, Method 514.3, Category 5 equipment, Procedure I, Test Conditions I-3.2.5. Pointer oscillation shall not exceed 2°C double amplitude and pointer variation shall not exceed 2°C. This test shall be conducted with power applied and with an input resistance that will provide an indication of 90°C.

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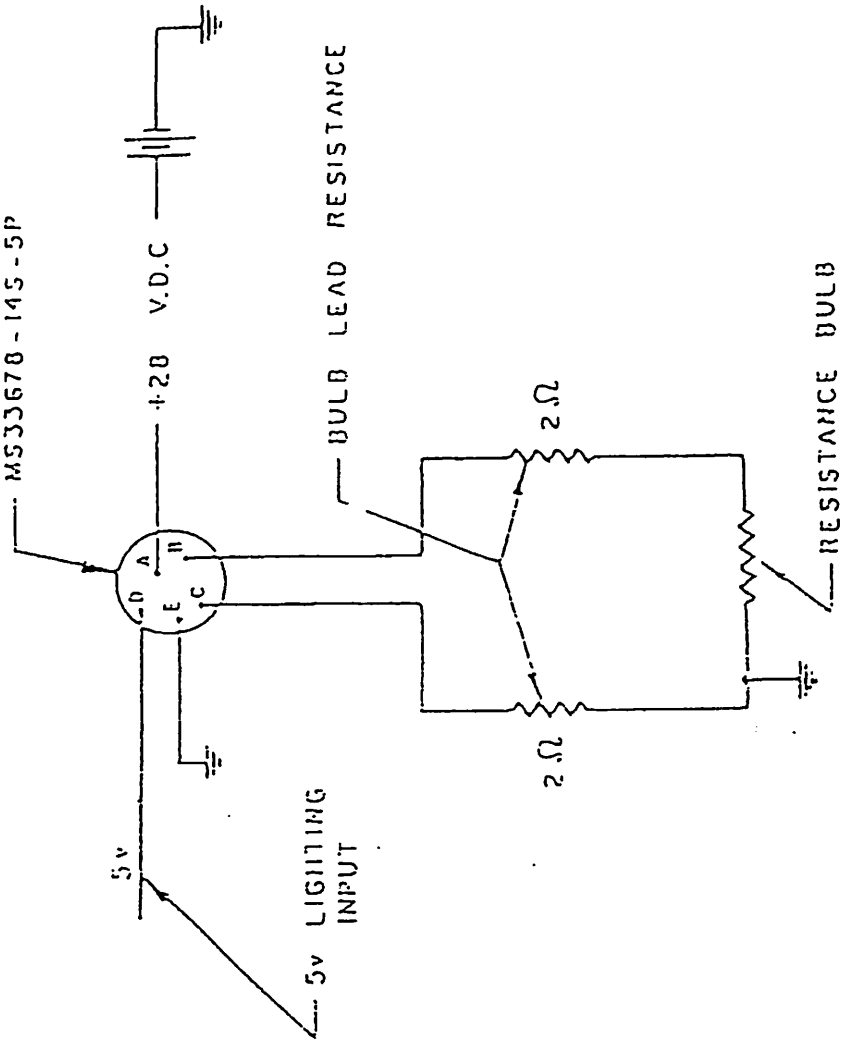


FIGURE 2. WIRING DIAGRAM

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4.4.2.8 Scale error at room temperature after vibration. After vibration, the scale errors shall not exceed the tolerances specified in Table I.

4.4.2.9 Sampling plan A lighting. The indicator shall be subjected to the sampling plan A tests of MIL-L-25467 or MIL-L-27160, as applicable.

4.4.3 Sampling plan B.

4.4.3.1 Response time. The response time shall not exceed 3 seconds.

4.4.3.2 Damping factor. The damping factor shall not be less than 1.6.

4.4.3.3 Power consumption. Power consumption shall not exceed 2 watts.

4.4.3.4 Bulb lead resistance. The indicator shall be connected as shown in figure 2 and subjected to the scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance by more than the tolerances specified in Table I.

4.4.3.5 Extreme low temperature exposure. After low temperature exposure, the scale errors at low temperature and the scale errors at room temperature shall not exceed the tolerances specified in Table I.

4.4.3.6 High temperature exposure. After high temperature exposure, the scale errors at room temperature shall not exceed the tolerances specified in Table I.

4.4.3.7 Salt spray. The indicators shall be subjected to and shall meet the sealing test following salt spray exposure. Insulation resistance shall be a minimum of 20 megohms.

4.4.3.8 Sampling plan B lighting. The indicator shall be subjected to the sampling plan B tests of MIL-L-25467 or MIL-L-27160, as applicable.

4.4.3.9 Thermal shock. At the completion of this test, the indicator shall be subjected to the test specified in 4.4.1.6. There shall be no evidence of moisture penetration or damage to the enclosure, and fogging of the cover glass during any part of this test shall be cause for rejection.

4.4.3.10 Reliability. The qualification test samples shall first be subjected to the scale error test and then to cycles consisting of 4 hours of operation with power and resistance applied and 1 hour with power removed. The cycles shall be conducted at the following ambient temperatures, ± 2°C, in the sequence specified: 4 hours at 0°C, 1 hour off; 4 hours at room temperature, 1 hour off; 4 hours at 50°C, 1 hour off. During the operational portion of the cycles, the resistance shall be so varied as to produce an indication cycle corresponding to a change from -60°C to +140° and return to -60°C. This shall constitute 1 indication cycle which shall be repeated at a rate of 6 ± 1 cpm unitl 75,000 indication cycles have been completed. The continuity of this test may be interrupted for weekend shutdown. At the conclusion of this test, the indicators shall be subjected to the scale error at room temperature and the

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friction error tests. The scale errors at the respective test points shall not exceed 150 percent of the scale errors prior to this test, and the friction error shall not exceed 3°C.

5. PACKAGING.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, indicators shall be individually preserved and packaged in accordance with MIL-P-116, method IC, and placed in a snug-fitting unit container conforming to PPP-B-676.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, or physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's standard practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping container shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.5.2 Level B. Indicators preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will ensure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

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5.4 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, _____*_____

*Applicable type designation to be added by the contractor.

5.4.1 Reinspection data markings. Reinspection data markings shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indicating temperatures, such as fuel, oil, and heat exchanger temperatures, ranging from -70° to +150°C.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of indicator required (see 1.2)

c. Levels of packaging and packing required and reinspection data markings (see section 5).

6.3 Definitions. Definitions will be in accordance with MIL-L-7070.

6.4 Subject terms (key words) listing.

-70° to 150°C
2-inch Diameter Face
Multifunction
Red Lighted
White Lighted

Custodian:
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Air Force - 82

(Project 6685-F853)

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